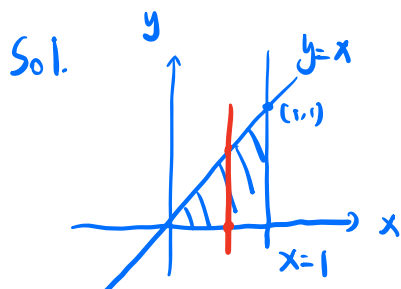


Lecture 12 Double Integrals Cont'd

Recap: Fubini's Theorem $\iint_R f(x,y) dA = \iint_R f(x,y) dx dy = \iint_R f(x,y) dy dx$

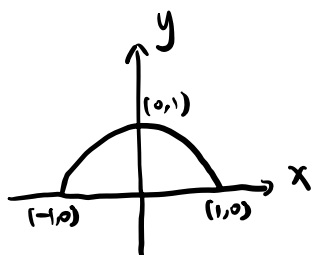
Example. Evaluate $\int_0^1 \int_y^1 \frac{\sin x}{x} dx dy$



$$\begin{aligned} & \int_0^1 \int_y^1 \frac{\sin x}{x} dx dy \\ &= \int_0^1 \int_0^x \frac{\sin x}{x} dy dx \\ &= \int_0^1 \left[y \frac{\sin x}{x} \right]_{y=0}^{y=x} dx \\ &= \int_0^1 \sin x dx \\ &= 1 - \cos 1 \end{aligned}$$

Example. Evaluate $\iint_R (x^2 + y^2) dA$

where R is the semicircular region bounded by the x -axis and the curve $y = \sqrt{1-x^2}$



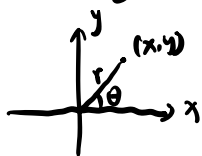
Sol.

$$\begin{aligned} & \int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx \\ &= \int_{-1}^1 \left(x^2 \sqrt{1-x^2} + \frac{(1-x^2)^{3/2}}{3} \right) dx \end{aligned}$$

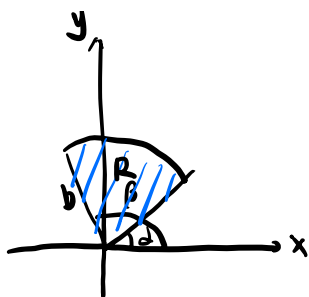
$R: 0 \leq r \leq 1, 0 \leq \theta \leq \pi$

$$\iint_R (x^2 + y^2) dA = \int_{\theta=0}^{\theta=\pi} \int_{r=0}^{r=1} r^2 r dr d\theta = \int_{\theta=0}^{\theta=\pi} \left[\frac{r^4}{4} \right]_{r=0}^{r=1} d\theta = \frac{\pi}{4}$$

Double Integrals in Polar Coordinates

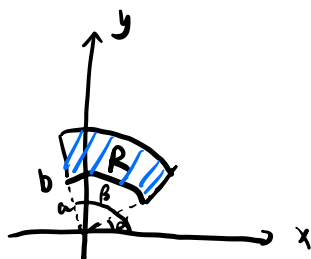


$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$



$$0 \leq r \leq b$$

$$\alpha \leq \theta \leq \beta$$



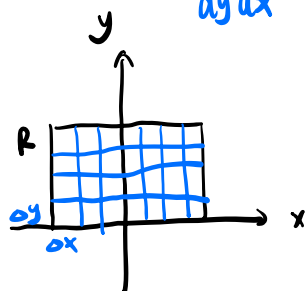
$$a \leq r \leq b$$

$$\alpha \leq \theta \leq \beta$$

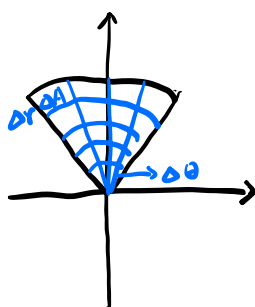
$$\iint_R f(x,y) \, dA = \iint_R f(r \cos \theta, r \sin \theta) \, dA = \iint_R f(r \cos \theta, r \sin \theta) r \, dr \, d\theta$$

~~$dr d\theta$ or $d\theta dr$?~~

$dA = r \, dr \, d\theta$ or $r \, d\theta \, dr$



$$\iint_R f(x,y) \, dA \approx \sum f(x,y) \Delta x \Delta y$$



$$A = \frac{1}{2} r^2 \theta$$

$$\Delta A = \frac{1}{2} (r + \Delta r)^2 \Delta \theta - \frac{1}{2} r^2 \Delta \theta$$

$$= r \Delta r \Delta \theta + \frac{1}{2} (\Delta r)^2 \Delta \theta$$

$$\Delta r, \Delta \theta \rightarrow 0$$

$$dA = r \, dr \, d\theta$$

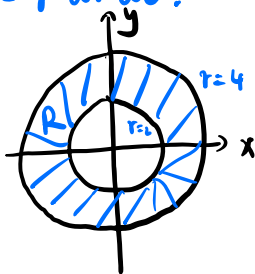
Example. Evaluate $\iint_R x \, dA$.

Sol. $R: 2 \leq r \leq 4, 0 \leq \theta \leq 2\pi$

$$\iint_R x \, dA = \int_0^{2\pi} \int_2^4 r \cos \theta \, r \, dr \, d\theta$$

$$= \left(\int_0^{2\pi} \cos \theta \, d\theta \right) \left(\int_2^4 r^2 \, dr \right)$$

$$= 0.$$



$$f(x,y) = g(x)h(y)$$

$$\iint_R f(x,y) \, dx \, dy$$

$$= \left(\int g(x) \, dx \right) \left(\int h(y) \, dy \right)$$

Change of variables for double integrals

(Chapter 14.7 of Calculus for Scientists and Engineers)

Given a transformation $T: x = g(u, v), y = h(u, v)$, where g and h are differentiable, the Jacobian (determinant) of T is

$$J(u, v) = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}.$$

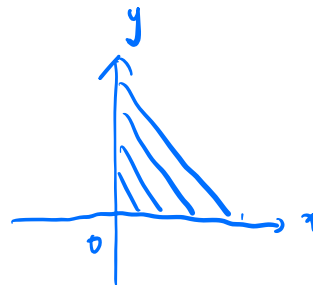
Then $\iint_R f(x, y) dx dy = \iint_R f(g(u, v), h(u, v)) |J(u, v)| du dv$.

$$J(r, \theta) = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

$$dx dy = |J(r, \theta)| dr d\theta = r dr d\theta.$$

Example Evaluate $\int_0^{+\infty} e^{-x} dx$.

$$\begin{aligned} \text{Sol. } & \int_0^{+\infty} \int_0^{+\infty} e^{-x^2-y^2} dx dy \\ &= \int_0^{+\infty} \int_0^{+\infty} e^{-x^2} \cdot e^{-y^2} dx dy \\ &= \left(\int_0^{+\infty} e^{-x^2} dx \right) \cdot \left(\int_0^{+\infty} e^{-y^2} dy \right) \\ &= \left[\int_0^{+\infty} e^{-x^2} dx \right]^2 \end{aligned}$$



$$\{R: x \in [0, +\infty), y \in [0, +\infty)\} = \{R: r \in [0, +\infty), \theta \in [0, \frac{\pi}{2}]\}$$

$$\begin{aligned} \int_0^{+\infty} \int_0^{+\infty} e^{-x^2-y^2} dx dy &= \int_0^{\pi/2} \int_0^{+\infty} e^{-r^2} r dr d\theta \\ &= \int_0^{\pi/2} d\theta \cdot \int_0^{+\infty} \frac{1}{2} e^{-r^2} d(r^2) \\ &= \pi/2 \cdot \left[-\frac{1}{2} e^{-r^2} \right]_{r=0}^{r=+\infty} \\ &= \frac{\pi}{4} \end{aligned}$$

$$\Rightarrow \int_0^{+\infty} e^{-x^2} dx = \sqrt{\frac{\pi}{4}} = \frac{\sqrt{\pi}}{2}$$

$$\int_{-\infty}^{+\infty} e^{-x^2} = \sqrt{\pi}$$

This is extremely important in probability,
particularly it is the probability density function of
normal distribution.